

Overview

This report documents the validation of both the Python and MATLAB implementations of Prandtl's Lifting-Line Theory (LLT) against classical analytical results. The solver is tested on an untwisted elliptic planform of aspect ratio $AR = 8$ using thin-airfoil input data ($c_l = 2\pi\alpha$, $c_d = 0$).

Validated at commit: e807286 on branch `integrated`.

Results apply to this specific version only; subsequent commits should be re-validated.

Lift Slope

For an elliptic wing in inviscid flow, classical lifting-line theory predicts a three-dimensional lift slope of

$$a = \frac{a_0}{1 + a_0/(\pi AR)}, \quad (1)$$

where $a_0 = 2\pi$ is the thin-airfoil sectional lift slope.

For $AR = 8$, equation (1) gives $a = 5.0265 \text{ rad}^{-1}$ (0.08773 deg^{-1}). Both solvers were run at angles of attack from -10° to $+10^\circ$ in 2° increments using 31 spanwise stations. The results are compared in Table 1.

Table 1: Comparison of lift slope from theory, Python, and MATLAB implementations (31 stations).

	Theoretical	Python	MATLAB
$dC_L/d\alpha \text{ (rad}^{-1}\text{)}$	5.0265	4.9957	4.9957
Error (%)	—	0.61	0.61

Both implementations produce identical results to six decimal places, with a relative error of 0.61% at 31 stations. This small discrepancy is attributable to the discrete approximation (31 spanwise stations, trapezoidal integration).

Induced Drag

For an elliptic wing, the induced drag coefficient is given exactly by

$$C_{D_i} = \frac{C_L^2}{\pi AR}. \quad (2)$$

Both solvers reproduce the theoretical drag polar to within machine precision. The maximum deviation between the computed C_D and $C_L^2/(\pi AR)$ is less than 6×10^{-5} for both implementations.

Convergence Study

A convergence study was performed to assess the effect of spanwise discretisation on the lift-slope error. Both solvers were run at five station counts spanning the range 31 to 1001.

The 1001-station case did not converge within 1000 iterations with the default relaxation factor of 0.01. This is a known limitation of the fixed-relaxation solver for very fine discretisations.

The Python and MATLAB results are identical at every station count, confirming that both implementations compute the same numerical solution.

Table 2: Convergence of lift slope error with increasing spanwise stations.

Stations	$dC_L/d\alpha$ (rad ⁻¹)		Error (%)
	Python	MATLAB	
31	4.995672	4.995672	0.614
151	5.018057	5.018057	0.169
301	5.021156	5.021156	0.107
601	5.022749	5.022749	0.076
1001	—	—	—

Summary

Both the Python and MATLAB lifting-line solvers have been validated against classical theory for an elliptic wing. The key results are:

- Python and MATLAB implementations produce **identical** numerical results (to six decimal places).
- Lift slope error: 0.61% at 31 stations, 0.076% at 601 stations.
- Induced drag follows the theoretical $C_{D_i} = C_L^2/(\pi AR)$ relation.
- Both solvers converge monotonically with increasing discretisation.

Both solvers are ready for use on arbitrary planforms and can accept either thin-airfoil theory or empirical airfoil data.